

AI Automated Podcast Transcript-Medi Lens

End-to-end AI system for podcast transcription and clinical analysis

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PROBLEM STATEMENT

Transforming Medical Audio Into Actionable Intelligence

Healthcare organizations face significant challenges in managing and analyzing audio content from clinical encounters, medical podcasts, and patient consultations.

Manual transcription is time-consuming, expensive, and prone to errors. Extracting meaningful insights, clinical summaries, and trending topics requires substantial human effort.

Key Challenges

- *Time-intensive manual transcription processes*
- *Difficulty extracting clinical insights at scale*
- *Limited ability to analyze sentiment and themes*
- *No standardized workflow for medical audio analysis*

Core Objectives



Accurate Transcription

Convert medical audio into precise, timestamped transcripts automatically using advanced ASR technology



Sentiment Analysis

Analyze emotional tone and sentiment across each audio segment for deeper understanding



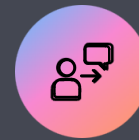
Keyword Extraction

Identify and visualize key medical terms, topics, and concepts through intelligent keyword clouds



Clinical Summaries

Generate automated, structured clinical summaries from medical dataset audio



Interactive Interface

Provide intuitive UI for uploading audio files and navigating segmented content seamlessly

Methodology



Audio Upload & Preprocessing

Audio files processed and optimized for transcription



ASR Transcription

OpenAI Whisper converts speech to accurate text



Topic Segmentation

Content divided into logical thematic segments



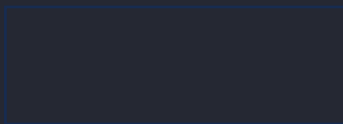
NLP Analysis

Sentiment, keywords, and clinical insights extracted



Output Generation

Summaries, visualizations, and interactive UI delivered



Tech Stack

Core Technologies

- *Python* - Primary language
- *Librosa & PyDub* - Audio processing
- *OpenAI Whisper* - Speech-to-text ASR

NLP & Analysis

- *NLTK & SpaCy* - Text processing
- *Hugging Face* - Transformer models
- *TF-IDF* - Keyword extraction

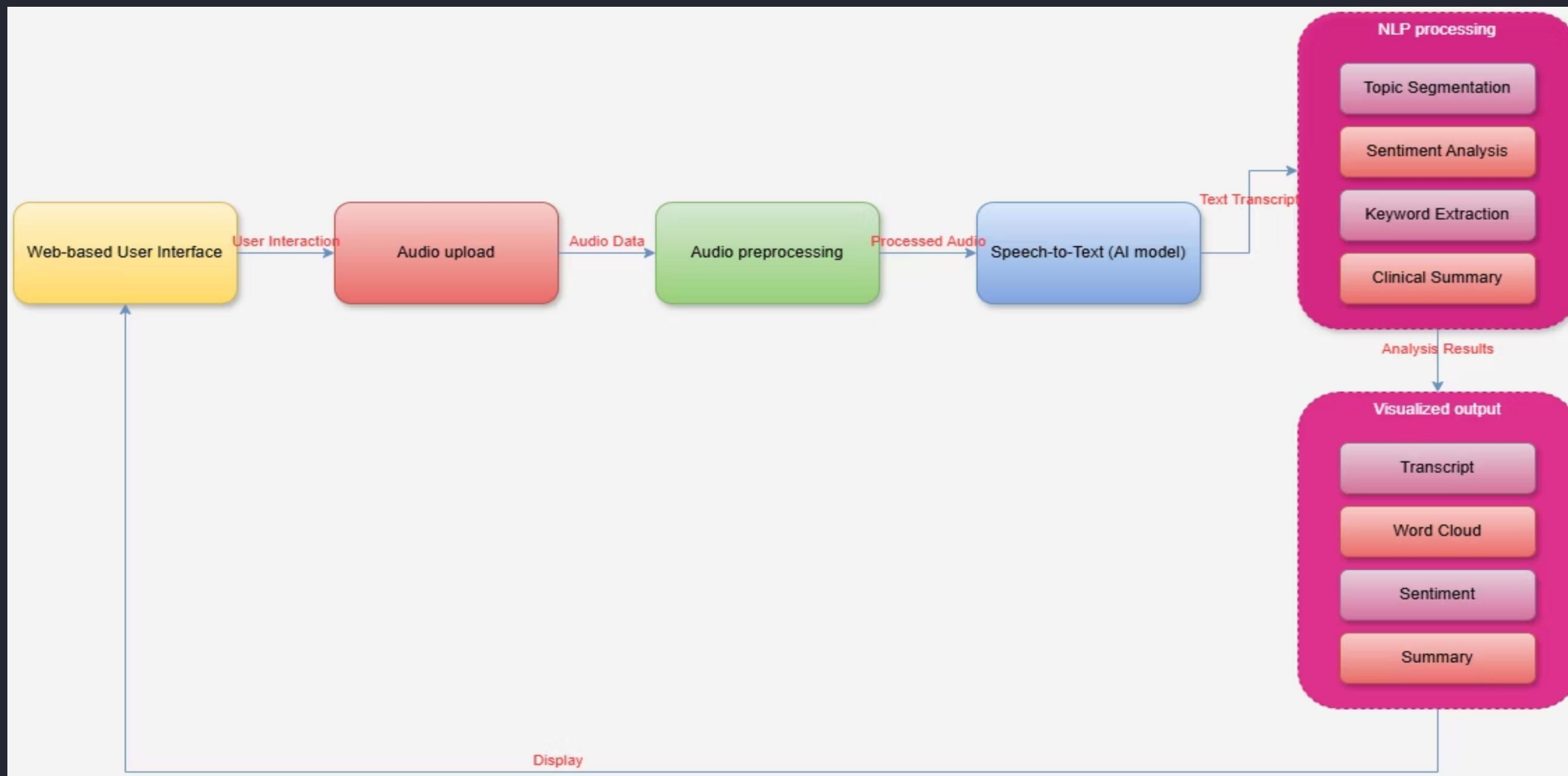
Visualization & UI

- *Streamlit* - Interactive interface
- *Plotly* - Data visualization
- *WordCloud* - Keyword visualization

```
def main():
    # Initialize the model
    model = load_model('model.h5')
    # Load the audio file
    audio = load_audio('audio.wav')
    # Process the audio
    spectrogram = librosa.amplitude_to_db(librosa.stft(audio))
    # Predict the sentiment
    prediction = model.predict(spectrogram)
    # Print the result
    print(prediction)

if __name__ == '__main__':
    main()
```

System Architecture



The architecture ensures efficient processing through modular components, enabling parallel analysis while maintaining data integrity throughout the pipeline.

Key Features & Capabilities



Automatic Audio Transcription

High-accuracy speech-to-text conversion with timestamp precision for medical terminology and clinical language



Segment-Level Sentiment Analysis

Granular emotional tone detection across conversation segments to identify patient concerns and clinical interactions



Keyword Cloud Visualization

Dynamic visual representation of key medical terms, conditions, and topics for rapid content understanding



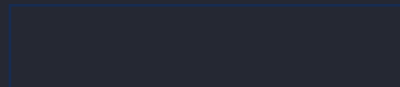
Clinical Summary Generation

AI-powered summarization of medical encounters with structured output following clinical documentation standards

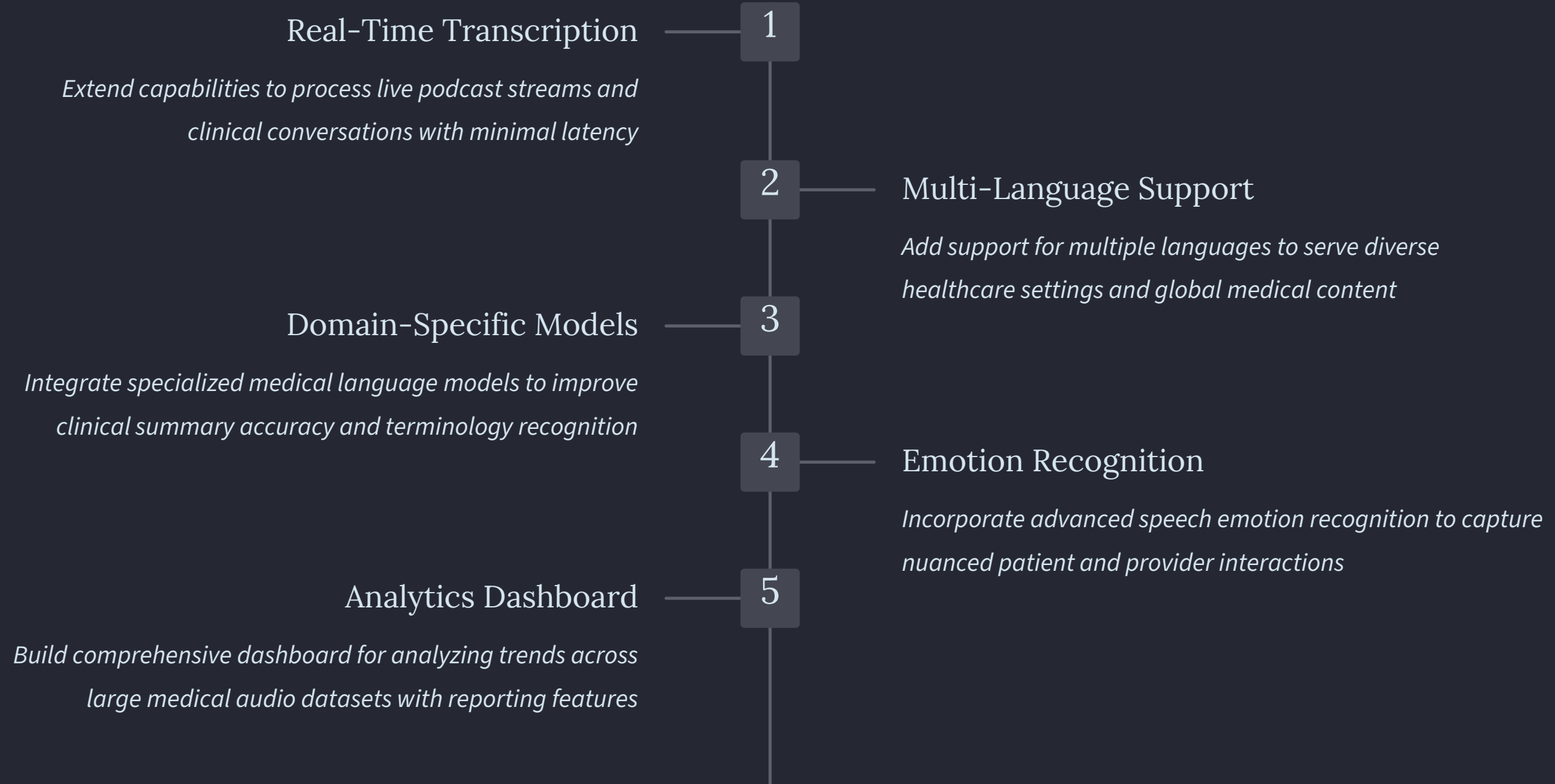


Interactive Streamlit UI

User-friendly interface for seamless audio upload, transcript navigation, and insight exploration

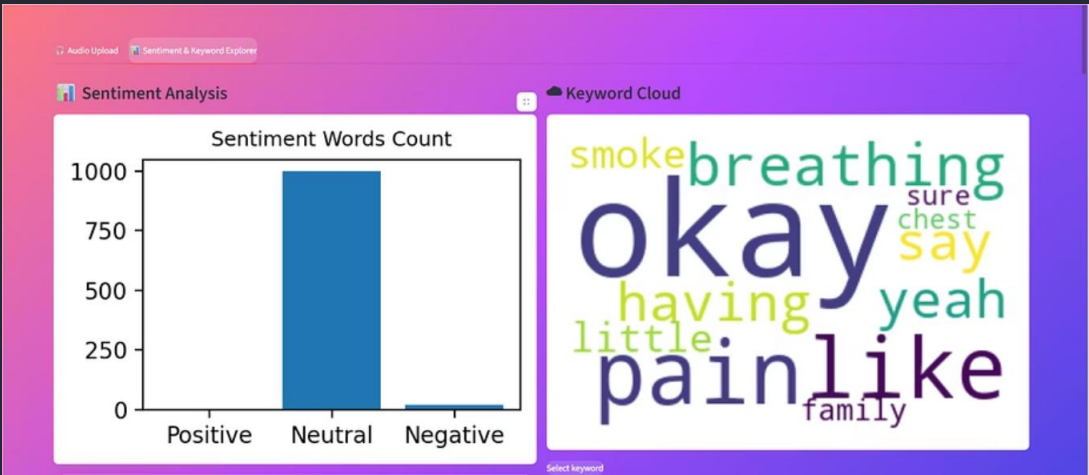
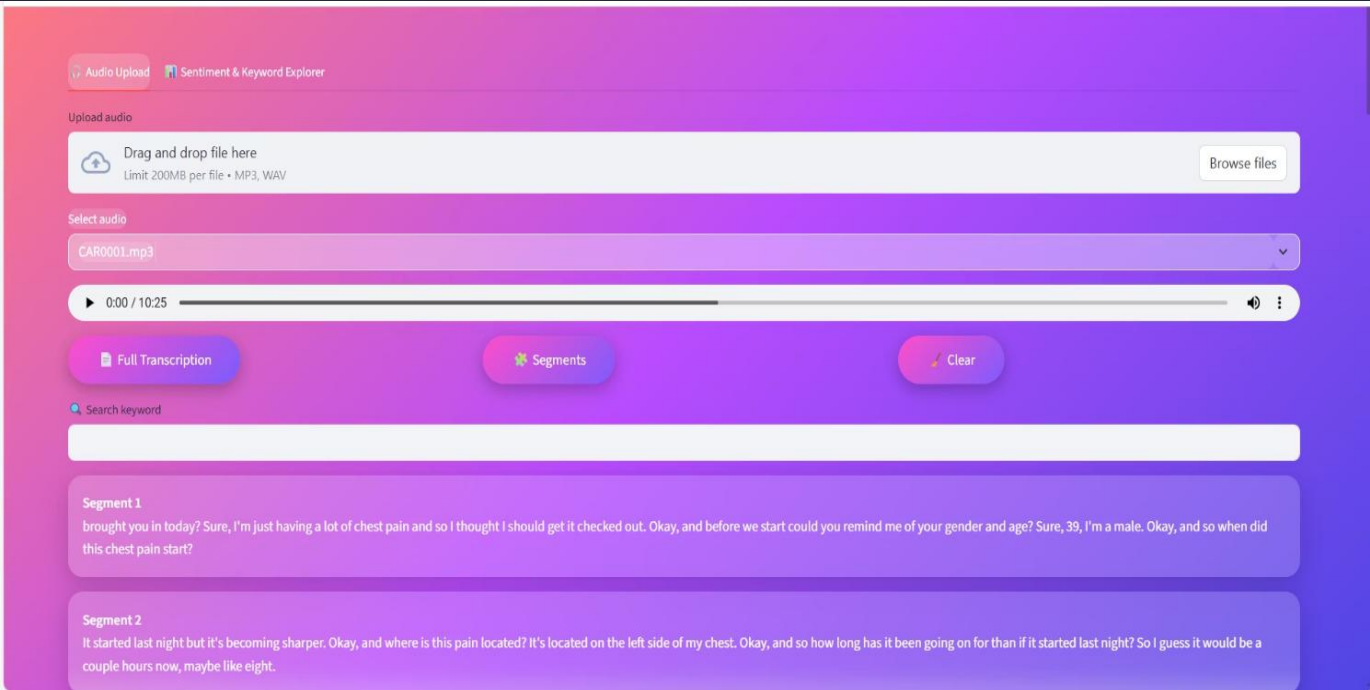


Future Scope & Enhancements



System Demonstrations

Interactive workflow showcasing the complete transcription and analysis pipeline



Positive Words

Neutral Words

maybe, they, brought, troubled, infectious, happened, hour, that, a, changing, moving, living

Negative Words

pain, severe

Select keyword

smoke

Keyword-based Segments

Okay, and how do you support yourself financially? I'm an accountant. Okay, sounds like a pretty stressful job, or that it can be. Do you smoke cigarettes? I do.

Okay, and how much do you smoke? I smoke about a pack a day. Okay, how long have you been smoking for? For the past 10 to 15 years. Okay, and do you smoke cannabis?

Sometimes. How much marijuana would you smoke per week? Per week? Maybe about 5 milligrams, not that much. Okay, and do you use any other recreational drugs like cocaine, crystal meth, opioids?

Auto Clinical Summary

OVERALL STATUS

HIGH RISK

POSITIVE INDICATORS

RISK FACTORS

pain

severe

PATIENT RECOMMENDATION

Immediate medical consultation is required.

Advantages



Time Savings

Automates transcription, drastically reducing time spent on manual processing.



Enhanced Accuracy

Ensures high precision and consistency, especially crucial for medical transcripts.



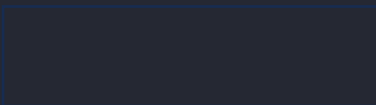
Cost Reduction

Eliminates the need for expensive manual transcription services.



Scalability

Effortlessly handles large volumes of audio content to meet growing demands.



Limitations

- ***Dependency on Audio Quality:** Performance can be significantly impacted by poor audio quality, background noise, or multiple speakers.*
- ***Limited Support for Specialized Medical Terminology:** While generally accurate, the system may struggle with highly specialized or obscure medical jargon.*
- ***Requires Internet Connectivity:** The transcription service relies on a stable internet connection, making offline use impossible.*
- ***Privacy and Data Security Considerations:** Handling sensitive medical information requires robust privacy protocols and adherence to data security regulations.*



Technical References & Resources

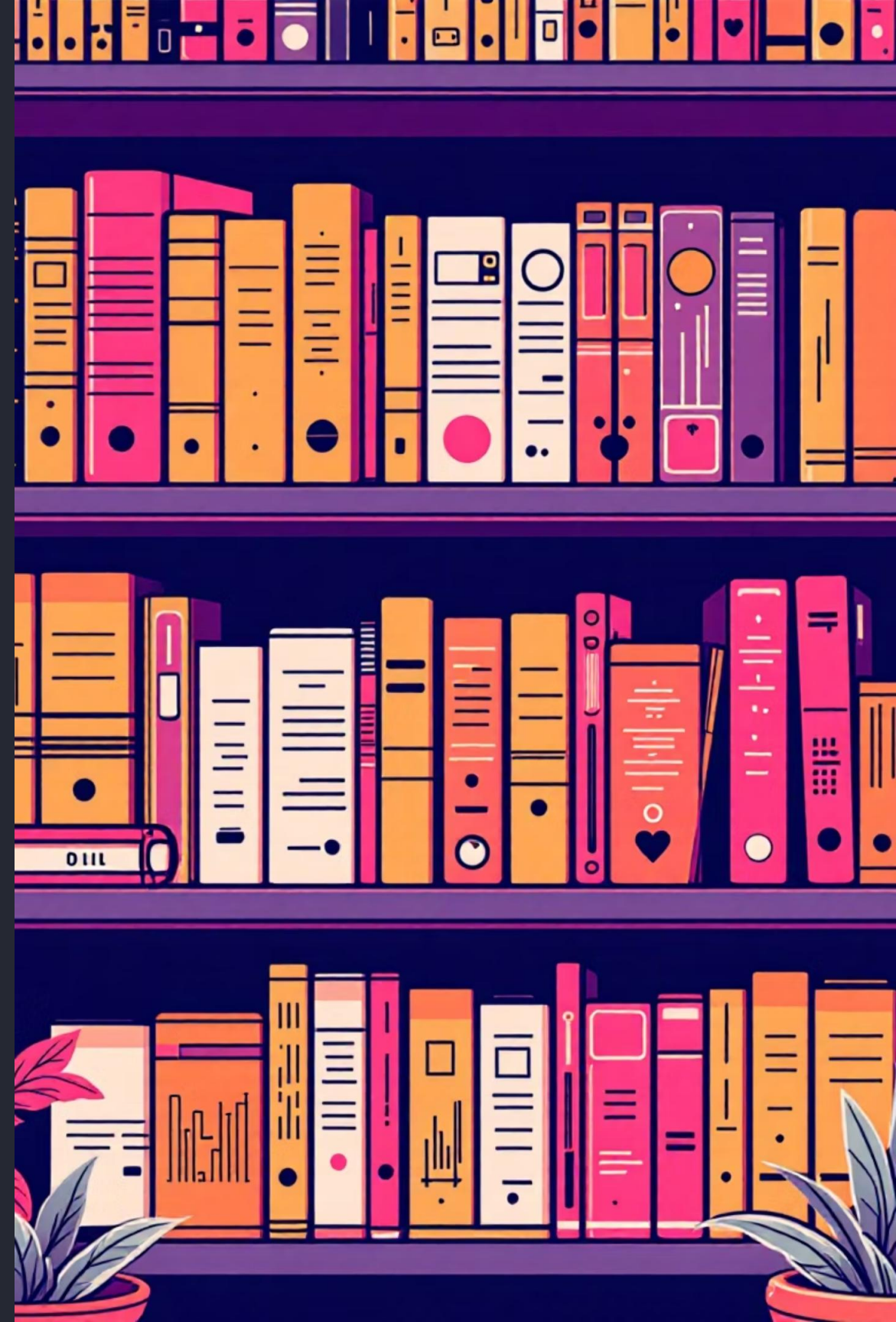
Core Libraries

- [OpenAI Whisper](#) - ASR transcription
- [Librosa](#) - Audio analysis
- [NLTK](#) - Natural language toolkit
- [Scikit-learn](#) - Machine learning

Visualization & UI

- [Streamlit](#) - Interactive UI framework
- [WordCloud](#) - Keyword visualization
- [Matplotlib](#) - Plotting library

📖 **Academic Reference:** *Speech & Language Processing* by Jurafsky & Martin - Stanford NLP Resource



Q&A

Thank you !!

